TAG COMPLETION WITH DEFECTIVE TAG ASSIGNMENTS VIA IMAGE-TAG RE-WEIGHTING

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ABSTRACT

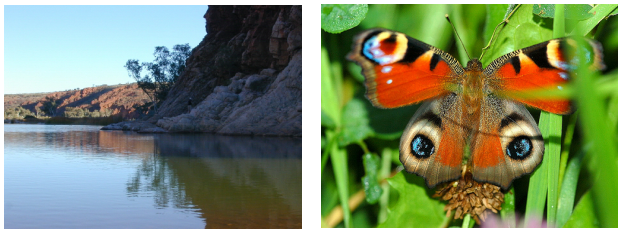
User-provided image tags are usually incomplete or noisy to describe the visual content of corresponding images. In this paper, we consider defective tagging which covers both incomplete and noisy situations, and address the problem of tag completion where tag assignments of training images are defective. While previous studies on tag completion usually assign equal penalty to empirical loss when processing each missing or noisy tag for each image, we show that this may be suboptimal as the relatedness of each tag to each image varies due to the defective setting. Thus, we introduce an image-tag re-weighting scheme to re-weight the penalty term of each tag to each image considering both image similarities and tag associations, and formulate a unified re-weighted empirical loss function. Experimental evaluations show that embedding proposed re-weighted empirical loss function in state-of-the-art tag completion algorithms achieves significant improvement in dealing with defective tag assignments.

***Index Terms*—** Tag completion, defection, image-tag re-weighting, image annotation

1. INTRODUCTION

In recent year, large quantities of images with user-provided textural tags have been posted to various social media on the web, and many exciting applications have been developed, e.g. tag-based image retrieval, image collection management. However, the raw user-provided tags tends to be ambiguous, incomplete even imprecise, which affects the practical usages of present applications. The coarse user-provided tags can be treated as *defective tag assignments*, which indicates that images are assigned with incomplete or noisy tags. As an example, an image whose true tag assignment is (t_1, t_2, t_3) is defectively assigned as (t_1) or (t_2, t_4). Fig. 1 shows examples of defectively tagged images from benchmark datasets IAPRTC-12 [1] and MIR Flickr [2].

To tackle the challenge of defection, our goal is to learn a tag completion model from defectively tagged training images to recover the complete tag assignments of training images, then apply the learnt model on untagged or defectively tagged test images. When applying on untagged images, our work is the same as image annotation models [3, 4, 5] which aims at assigning multiple tags to unlabeled images. It is worth noting that these annotation models assumes training images are completely tagged without miss or noise. However, the defective settings could potentially result in biased annotation result, thus it is suboptimal to directly utilize these traditional image annotation methods.



cliff, lake, landscape, shade, sky, ***pond****,* ***rock****,* ***tree****,* ***water***

nature, macro, flowers, germany, animal, eyes, insect, ***plant****,* ***leaf****,* ***wildlife***

Fig. 1. Example images with defective tag assignments in datasets IAPRTC-12 (left) and MIR Flickr (right): italic tags are provided ground truth tags, underlined tags are incorrect noisy tags, and bold tags are potentially missing tags.

When applying on defectively tagged test images, our work is related to tag completion on partially tagged images and tag refinement on tag denoising. The former considers tag completion with clean but partial manual annotations. And the ongoing researches mainly extend conventional multi-label ranking [6], ridge regression [7] or binary SVM classification [4] approaches by exploiting the potential associations between the assigned and unassigned tags. Furthermore, recently studies [8, 9] focus on matrix factorization (reconstruction) methodologies by embedding various contextual information such as semantic and visual similarities. The latter performs tag refinement to differentiate noisy tags from correct tags using tag relevance [10, 11], denoising sparse error [12], collaborative propagation among tags [13], etc.

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Our work can be treated as unified framework that tackles both tag completion and denoising. Although several previous studies [8, 13, 12] are also declared to be unified frameworks, an inevitable disadvantage of them is the difficulty in balancing the degree of completion and denoising. When recalling missing tags (completion) or rejecting noisy tags (denoising) from an image, the penalties of different tags are equally allocated, leading the result to be biased on either completion or denoising. This potential disadvantage motivates us to consider specific image-tag penalty term when assigning each tag to each image. Imaging the examples in Fig. 1, the cost of recalling tag "*pond*" ought to be different from tag "*car*," as well the penalty of rejecting "*germany*" would vary from "*animal*."

The main contributions of our research are summarized as follows: 1)We propose an efficient re-weighting scheme to elaborately assign the penalty of predicting each tag to each image, and the penalty of pairwise image-tag is fully derived from both image-image similarities and tag-tag associations. 2)We formulate a unified re-weighted empirical loss term utilizing the re-weighted penalty term. 3)We extend two recently proposed tag completion models to incorporate the re-weighted empirical loss term, developing the corresponding optimization algorithms to perform tag completion with defective tag assignments.

2. IMAGE-TAG RE-WEIGHTING

We first describe the proposed re-weighted empirical loss function and then present an efficient scheme to determine the image-tag weights combining both visual and semantic cues.

2.1. Re-weighted empirical loss

Let $\mathcal{X} = \{x_1, ..., x_n\}$ be a collection of n images and $\mathcal{Y} = \{t_1, ..., t_m\}$ be a vocabulary of m tags. The user-annotated dataset $\hat{T}_{n\times m} = \{\hat{Y}_1, ..., \hat{Y}_n\}$ is a defective tag matrix, where $\hat{Y}_i \subseteq \mathcal{Y}$ is an initial defective tag vector of image x_i, and $\hat{Y}_{i,j}$ is set to one if tag j is assigned to image i and zero otherwise. Our target is to recover the final complete tag matrix $T_{n\times m}$ from initial defective $\hat{T}$, where the element $T_{i,j} \geq 0$ denotes confidence score of assigning tag t_j to image x_i. The general formulation of previous tag completion works [7, 8, 12, 13] corresponds to an empirical loss minimization with a regularization term as follows:

$$\min_T \lambda E_p(T) + E_r(T), \tag{1}$$

where the regularization term $E_p(T)$ restricts the model to some constraints, such as the visual and semantic consistencies, and $\lambda \geq 0$ controls the trade-off between empirical loss and regularization term. Specifically, the empirical loss term $E_r(T)$ represents the deviation of complete tag matrix T from initial defective tag matrix $\hat{T}$, and is generally formulated as

$$E_r(T) = \|T - \boldsymbol{\alpha}\hat{T}\|_F^2 = \sum_{i=1}^{n}\sum_{j=1}^{m}(Y_{i,j} - \alpha_{i(j)}\hat{Y}_{i,j})^2. \tag{2}$$

Here $\|\cdot\|_F$ denotes Frobenius matrix norm, $\boldsymbol{\alpha}$ is an $n \times n$ diagonal matrix where α_i plays as scaling factor for image x_i. Please note $\boldsymbol{\alpha}$ could also be an $m \times m$ diagonal matrix, where α_j is a scaling factor for tag t_j. With respect to the selection of $\boldsymbol{\alpha}$, previous tag completion works can be categorized into two types: 1) **Identical-type**, which assumes $\boldsymbol{\alpha}$ to be an identical diagonal matrix. This type indicates that each element in $\hat{T}$ enjoys identical value. 2) **Scaling-type**, which assumes $\boldsymbol{\alpha}$ to be a diagonal matrix with different diagonal values. This type implies that each row (column) in $\hat{T}$ has equal value α_i (α_j).

It is indicated from Eq. 2 that: 1) the complete tag matrix T should be consistent (similar) to the initial defective matrix $\hat{T}$, 2) the penalty of assigning each tag t_j to each image x_i contributes equally to empirical loss $E_r(T)$. However, these two assumptions may be inappropriate to obtain the optimal T under defective setting. Firstly, given the defective tag matrix $\hat{T}$, the constraint of consistency between $\hat{T}$ and T is too rigorous to twist the complete T from defective $\hat{T}$, which leads the final T to be similar with original $\hat{T}$ (see more interpretation in Fig.2). Secondly, the identical penalty assumption of assigning each tag t_j to each image x_i is imprecise and inhibits the model from recalling missing tags and rejecting noisy tags from $\hat{T}$.

To formulate a more appropriate empirical loss $E_r(T)$ under defective tag assignments, we introduce a weighted image-tag penalty matrix $Z_{n\times m}$, whose element $Z_{i,j}$ represents the penalty of assigning tag j to image i in T. Then, the re-weighted empirical loss can be derived as:

$$E_r(T) = \|(T - \boldsymbol{\alpha}\hat{T}) \circ Z\|_F^2 = \sum_{i=1}^{n}\sum_{j=1}^{m}(Y_{i,j} - \alpha_{i(j)}\hat{Y}_{i,j})^2 Z_{i,j}. \tag{3}$$

Here, $\circ$ represents point-wise product of matrices. It is worth noting that the specific image-tag penalty $Z_{i,j}$ in Z is determined by corresponding image-tag relatedness. Specifically, for tag t_j that is related but missing in image x_i, the penalty $Z_{i,j}$ needs to be smaller that higher confidence score can be allocated to recall t_j in x_i. Likewise, for tag t_j that is noise but tagged in image x_i, the penalty $Z_{i,j}$ ought to be smaller that lower confidence score can be allocated to reject t_j in x_i.

Compared with traditional empirical loss in Eq.2, the re-weighted empirical loss in Eq. 3 indeed have several advantages: 1) It allows the complete tag matrix T to be moderately inconsistent to the initial defective matrix $\hat{T}$, and it is possible to recall untagged related tags which are missing in $\hat{T}$ and remove tagged noisy tags from $\hat{T}$, since the weighted penalty Z could balance the empirical loss. 2) Each element $Z_{i,j}$ in

Z is flexible to explicitly incorporate both visual and semantic cues to model the specific image-tag relatedness, while the identical penalty in Eq. 2 would neglect the discrimination of pairwise image-tag and limitedly formulate a suboptimal empirical loss term. In Fig. 2, we present an illustration of

Groud-truth tags:
nature, macro, animal, insect, plant, leaf, wildlife

Defective tags $\hat{T}$:
nature, germany, eyes, insect

Complete tags without re-weighting T:

nature (1)(0.9)	insert (1)(0.9)	eyes (1)(0.7)
animal (1)(0.5)	germany (1)(0.4)	

Complete tags with re-weighting T:

nature (1)(0.9)	insert (1)(0.8)	~~eyes~~ (0.3)(0.2)
~~germany~~ (0.3)(0.1)	animal (0.2)(0.6)	wildlife (0.3)(0.5)
plant (0.3)(0.4)	leaf (0.3)(0.4)	

Fig. 2. Tag completion example with/without image-tag re-weighting: $\hat{T}$ is initial defective tags for training, T is final tag completion result. Tags in T is in the format of "tag ($Z_{i,j}$)(confidence score)." The empirical loss without re-weighting (Eq.2) is: $0.1^2 + 0.1^2 + 0.3^2 + 0.5^2 + 0.6^2 = 0.72$, with re-weighting (Eq.3) is: $0.1^2 + 0.2^2 + 0.8^2 \times 0.3 + 0.9^2 \times 0.3 + 0.6^2 \times 0.2 + 0.5^2 \times 0.3 + 0.4^2 \times 0.3 + 0.3^2 \times 0.4 = 0.716$. Note that the rejected noisy tags "*eyes*," "*germany*" also contribute to the re-weighted loss in Eq.3.

tag completion results with/without image-tag re-weighting in identical-type model ($\boldsymbol{\alpha}$ is identical diagonal matrix). With similar empirical loss values (0.72 *vs.* 0.716), the recovered tag matrix T without re-weighting (top box in Fig. 2) twists initial $\hat{T}$ slightly, since using identical penalty it is hard to decline the confidence scores of noisy tags (e.g. *eyes*, *germany*) and promote missing tags (e.g. *plant*, *leaf*) while keeping low empirical loss. However, after importing the re-weighted image-tag penalties, T with re-weighting (bottom box in Fig.2) is able to assign appropriate confidence scores relying on image-tag relatedness while maintaining low empirical loss value.

2.2. Determining image-tag penalty

We propose an efficient approach to determining $Z_{i,j}$ for pairwise image-tag entry in $\hat{T}$ incorporating both visual and semantic similarities. For a given tag $t_j, \forall j \in \{1, ..., m\}$, let $\mathcal{X}_{t_j}^+$ be the set of images that are annotated with tag t_j, and the remaining samples as $\mathcal{X}_{t_j}^- = \mathcal{X} \backslash \mathcal{X}_{t_j}$, and we consider $Z_{i,j}$ for $x_i \in \mathcal{X}_{t_j}^+$ and $x_i \in \mathcal{X}_{t_j}^-, i \in \{1, ..., n\}$ separately.

- Visual similarity based relatedness $R_V(\cdot)$. We compute the visual distance score $dist(\cdot)$ (scaled to range $[0, 1]$) of x_i with all images $x_k \in \mathcal{X}_{t_j}$. Specially, for $x_i \in \mathcal{X}_{t_j}^+$, $R_V(x_i, t_j) = 1 - \min dist(x_i, x_k), x_k \in \mathcal{X}_j^+ \backslash x_k$, for $x_i \in \mathcal{X}_{t_j}^-$, $R_V(x_i, t_j) = 1 - \min dist(x_i, x_k), x_k \in \mathcal{X}_{t_j}^+$.
- Semantic similarity based relatedness $R_S(\cdot)$. We first compute the correlation score between pairwise tags t_i and t_j as: $co_occur(t_i, t_j) = \frac{f_{i,j}}{f_i + f_j - f_{i,j}}$, where f_i and f_j are the occurrence of tags t_i and t_j, and $f_{i,j}$ is the co-occurrence of tags t_i and t_j. Let $\hat{Y}_i$ be the tag set of image x_i, we define $R_S(x_i, t_j) = \max_{t_k \in \hat{Y}_i} co_occur(t_j, t_k)$.

Based on the two similarity measures above, we define the relatedness $R(x_i, t_j)$ as a weighted linear combination of the two similarity components above, as:

$$R(x_i, t_j) = \pi R_V(x_i, t_j) + (1 - \pi) R_S(x_i, t_j), \qquad (4)$$

where $R_V(\cdot)$ and $R_S(\cdot)$ are normalized similarity scores from image-specific and tag-specific measures, and π is a weight parameter in $[0, 1]$ which is validated during parameter turning. For each tag t_j, we calculate the image-tag penalty $Z_{i,j}$ for the i-th image based on $R(x_i, t_j)$ as:

$$Z_{i,j} = \begin{cases} R(x_i, t_j) & \text{if } x_i \in \mathcal{X}_{t_j}^+ \\ \frac{1}{1+\exp(R(x_i, t_j))} & \text{if } x_i \in \mathcal{X}_{t_j}^- \end{cases} \qquad (5)$$

From Eq.5 it can be learnt that missing and noisy tags have smaller penalties $Z_{i,j}$ than other tags depending on their relatedness value $R(x_i, t_j)$. Thus these tags are more likely to be predicted with higher confidence scores, and they are able to be recalled (missing tags) or rejected (noisy tags) in recovered complete tag matrix T.

3. TAG COMPLETION VIA IMAGE-TAG RE-WEIGHTING

In this section, we select two typical tag completion models: tag matrix completion (TMC) [8] from identical-type models and Fasttag [7] from scaling-type models, embed proposed image-tag re-weighting scheme and derive the parameter optimization algorithms for these two types of models.

3.1. TMC with image-tag re-weighting

Given the visual features of image data $V_{n \times d}$, tag correlation matrix $R_{m \times m}$ and observed defective tag matrix $\hat{T}_{n \times m}$, the TMC model reconstructs the complete tag matrix $T_{n \times m}$ under three constraints: (1) the complete tag matrix T should be similar to the observed defective matrix $\hat{T}$, and the empirical loss term is formulated as $\|T - \hat{T}\|_F^2$, (2) the complete tag matrix T should be consistent with the visual similarity, this constraint is formulated as $\|TT^\top - V diag(w) V^\top\|_F^2$, where $w_{d \times 1}$ is a non-negative vector representing the weight of each visual feature dimension, (3) the complete matrix T

should also be consistent with tag correlation matrix R, it can be formulated as $\|TT^\top - R\|_F^2$. It is indicated that TMC is an identical-type model, as $\boldsymbol{\alpha}$ is set to be an identical diagonal matrix in the first constraint.

When embedding proposed re-weighting scheme in TMC model, we replace the first constraint $\|T - \hat{T}\|_F^2$ with the re-weighted empirical loss term in Eq.3. Thus, the objective of TMC model with image-tag re-weighting (TMC-RW) is to minimize the loss function $\mathcal{L}(T, w)$, which is derived as,

$$\min_{T,w} \mathcal{L}(T,w) = \min_{T,w} \underbrace{\|(T - \hat{T}) \circ Z\|_F^2}_{re-weighted\ loss} + \eta\|TT^\top - R\|_F^2 + \lambda\|TT^\top - Vdiag(w)V^\top\|_F^2 + \mu\|T\|_1 + \gamma\|w\|_1, \quad (6)$$

where the $L1$ regularizer for both T and w are to generate sparse solutions for them. We adopt the subgradient descent based algorithm proposed in [8] to optimize T and w with non-negative constraints. Define $\mathcal{L}(T,w) = \mathcal{A}(T,w) + \mu\|T\|_1 + \gamma\|w\|_1$, $G = TT^\top - Vdiag(w)V^\top$, $H = TT^\top - R$, the subgradients for $\mathcal{A}(T,w)$ are derived as

$$\begin{aligned} \bigtriangledown_T \mathcal{A}(T,w) &= 2\lambda GT + 2\eta TH + 2(T - \hat{T}) \circ Z, \\ \bigtriangledown_w \mathcal{A}(T,w) &= -2\lambda diag(V^\top GV), \end{aligned} \quad (7)$$

for $L1$ regularizer and non-negative constraints to T and w, we follow the composite function optimization used in [8], and derive the final update rules for T and w in each iteration:

$$\begin{aligned} T &\leftarrow [T - \zeta_t \bigtriangledown_T \mathcal{A}(T,w) - \mu\zeta_t \mathbf{1}_{n\times m}]_+, \\ w &\leftarrow [w - \zeta_t \bigtriangledown_w \mathcal{A}(T,w) - \gamma\zeta_t \mathbf{1}_{d\times 1}]_+, \end{aligned} \quad (8)$$

where ζ_t is the step size and $[f]_+ = \max(0, f)$ projects the negative entries in T and w to be zero in iteration t.

3.2. Fasttag with image-tag re-weighting

To keep the notations consistent with Fasttag, we denote the visual features of images data $V_{d\times n} = \{x_1, ..., x_n\}$ and observed defective tag matrix $\hat{T}_{m\times n} = \{\hat{Y}_1, ..., \hat{Y}_n\}$ for each image in each *column*. Fasttag assumes that the observed defective tag matrix $\hat{T}$ is a "corrupted" version of the complete tag matrix $T_{m\times n}$, and introduces an enrichment mapping matrix $B_{m\times m}$ to reconstruct T from $B\hat{T}$ by estimating which tags are likely to be recalled or rejected in $\hat{T}$. The original empirical loss is formulated as $\|T - B\hat{T}\|_F^2$, which implies that Fasttag is a scaling-type model as each image has identical value of $\boldsymbol{\alpha}$. Moreover, besides the enriched mapping to reconstruct complete tag matrix T, T can also be reconstructed from image features V. Then a co-regularizer term is introduced to force the two reconstructions from both tag-side and image-side with small difference $\|B\hat{T} - WV\|_F^2$, where each row of $W_{m\times d}$ contains the weights of a linear classifiers for each tag. After embedding the re-weighted empirical loss term, the objective of Fasttag with image-tag re-weighting (Fasttag-RW) is to minimize the loss function $\mathcal{L}(W, B)$ as

$$\min_{B,W} \underbrace{\|(T - B\hat{T}) \circ Z\|_F^2}_{re-weighted\ loss} + \gamma\|B\hat{T} - WV\|_F^2 + \lambda\|W\|^2, \quad (9)$$

where $L2$ regularization on W is to reduce complexity and avoid overfitting. In Eq. 9, we do not directly optimize T, instead we use the backtracked corruption method [7] to approximate the re-weighted empirical loss. In particular, we randomly remove some tags with independent probability $p \in (0,1)$ from $\hat{T}$ to generate a further corrupted version $\bar{T} = \{\bar{Y}_1, ..., \bar{Y}_n\}$ from $\hat{T}$, and we repeat this backtracked corruption procedures several times, and use the expected corruption value to approximate re-weighted empirical loss. Finally, the empirical loss can be approximated as

$$\begin{aligned} \|(T - B\hat{T}) \circ Z\|_F^2 &\approx \mathbb{E}\left[\|(\hat{T} - B\bar{T}) \circ Z\|_F^2\right]_{p(\bar{T}|\hat{T})} \\ &= \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}\left[\|(\hat{Y}_i - B\bar{Y}_i) \circ \boldsymbol{z}_i\|_F^2\right]_{p(\bar{Y}_i|\hat{Y}_i)}, \end{aligned} \quad (10)$$

where $\boldsymbol{z}_i$ is the i-th column in Z for the i-th image. When expanding Eq.10, we define $P = \sum_{i=1}^{n} \hat{Y}_i \mathbb{E}[\bar{Y}_i \circ \boldsymbol{z}_i]^\top$ and $Q = \sum_{i=1}^{n} \mathbb{E}[\bar{Y}_i \circ \boldsymbol{z}_i \bar{Y}_i^\top]$, then the re-weighted empirical loss in Eq.10 can be rewritten as

$$\|(T - B\hat{T}) \circ Z\|_F^2 \approx \frac{1}{n} trace(BQB^\top - 2PB^\top + \hat{T} \circ Z\hat{T}^\top). \quad (11)$$

Here P and Q can be computed as

$$\begin{aligned} P &= (1-p)\hat{T} \circ Z\hat{T}^\top, \\ Q &= (1-p)^2 \hat{T} \circ Z\hat{T}^\top + p(1-p)\delta(\hat{T} \circ Z\hat{T}^\top), \end{aligned} \quad (12)$$

where $\delta(\cdot)$ is an operator to set all entries except the diagonal to zero. By substituting Eq. 11 and Eq. 12 in Eq. 9, we can use block-coordinate descent algorithm to optimize B and W alternatingly. Finally, the close form solutions for B and W are derived as

$$\begin{aligned} B &= (P + \gamma WV\hat{T}^\top)(Q + \gamma\hat{T}\hat{T}^\top)^{-1}, \\ W &= \gamma B\hat{T}V^\top(\gamma VV^\top + n\lambda \mathbf{1}_{d\times d})^{-1}. \end{aligned} \quad (13)$$

Due to the space of limitation, we have omitted deriving details for parameter optimization in proposed two re-weighted based models above. We will supply a technical note with this paper online for completeness. It is worth noting that proposed two re-weighted models TMC-RW and Fasttag-RW maintain the same level of computational complexity since the close-form solutions for parameters in Eq.8 and Eq.13 have the same forms as TMC and Fasttag models and only the products of matrices are changed involving in the re-weighted penalty Z. In practice, Z can be calculated off-line from the initial defective tag matrix $\hat{T}$ and fixed during each iteration.

4. EXPERIMENTS

4.1. Experimental configuration

Datasets and Features. We use three benchmark datasets ESP Game [14], IAPRTC-12 [1] and MIR Flickr [2] in our experiments, which are very challenging with large tag vocabulary and significant diversity among visual content. Table 1 shows general statistics of these datasets. The original multiple features including global descriptors (RGB, HSV, LAB, Gist) and local descriptors (SIFT, Hue) were extracted in [5], and widely utilized in subsequential works [4, 6, 7]. Here we adopt the multiple feature mapping and random projection scheme in [7], separately perform dimensionality reduction for all original features, and concatenate to a 12912-dimensional merged feature to represent each image.

Table 1. General statistics of three datasets. In column 2, counts of images are in the format "training, test." In column 4 and 5, the counts are in the format "mean, minimum, maximum."

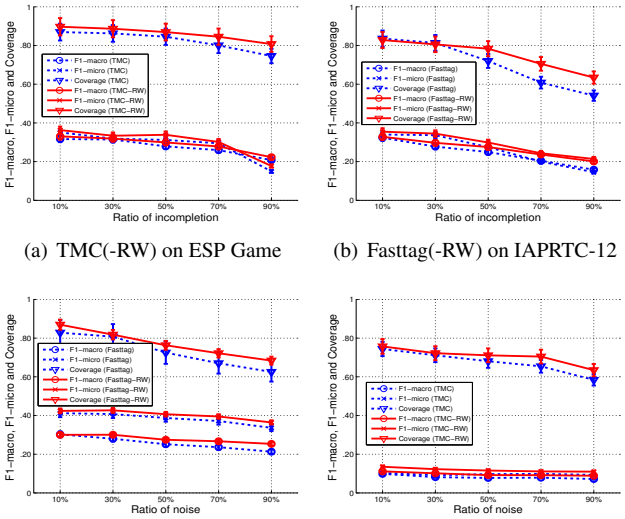
Dataset	Images	Labels	Labels per image	Images per label
ESP Game	18689, 2081	268	4.7, 1, 15	326, 172, 4553
IAPRTC-12	17665, 1962	291	5.7, 1, 23	34, 153, 4999
MIR Flickr	9359, 9335	457	4.55, 1, 45	145, 50, 1483

Defective tag assignments. To simulate the situation of defective tag assignments, we conduct experiments in two settings: incomplete setting and noisy setting. (1) For incomplete setting, we randomly delete partial tags for each images , the deletion process stands by the principle $\min(1, \lceil N \times (1 - ratio) \rceil)$ ensuring that each images preserves at least one tag. (2) For noisy setting, we randomly add untagged tags for each images, the addition process follows the principle $N + \min(1, \lceil N \times ratio \rceil)$, ensuring that each image is corrupted by at least one noisy tag. Here N denotes the counts of originally tagged tags in one image, $\lceil \cdot \rceil$ denotes the ceiling function which gives the smallest integer not smaller than the given value, $ratio$ represents the degree of incompletion or noise. In our experiments, we set $ratio = \{10\%, 30\%, 50\%, 70\%, 90\%\}$. It indicates that the larger the ratio, the higher the degree of incompletion or noise.

Evaluation metrics. We synthesize various evaluation criteria used in previous tag completion works [6, 7, 8] and tag refinement works [10, 11], finally adopt three criteria: (1) *F1-macro*, which is the averaged F1 score ($F1 = 2\frac{Precision * Recall}{Precision + Recall}$) of all tags, (2) *F1-micro*, which is the averaged F1 score of all images, (3) *Coverage* ($Coverage = \frac{N_+}{N_{total}}$), where N_+ is the count of recalled tags by model and N_{total} is the quantity of tags in vocabulary. To calculate these criteria, all images are predicted with top five tags which have highest confidence score. Notable that for all three criteria, larger numerical value indicates better performance.

4.2. Evaluation on various degree of defection

We first compare proposed re-weighting based TMC-RW and Fast-RW models with previous TMC and Fasttag on various degree of incomplete and noisy settings. To obtain the best parameters for these models, we randomly split 30% initial training samples in each dataset as validation set, and perform model selection. We repeat experiments five times for each model, and report the evaluation results averaged over the five trials.



(a) TMC(-RW) on ESP Game (b) Fasttag(-RW) on IAPRTC-12

(c) Fasttag(-RW) on IAPRTC-12 (d) TMC(-RW) on MIR Flickr

Fig. 3. Tag completion performance of proposed methods with various degree of incomplete and noisy settings.

Fig. 3 compares tag completion performance of proposed re-weighting based models and previous models without re-weighting scheme. It can be observed that in most conditions, varying the degree of incomplete and noisy settings, tag completion results are constantly improved on all three evaluation metrics when re-weighting scheme is taken into account. In particular, both TMC-RW and Fasttag-RW models are robust to serious condition of defection. For degree of incompletion and noise from 50% to 90%, TMC-RW and Fasttag-RW drop slightly on three measures while TMC and Fasttag decrease drastically. The reason behind this is that when the degree of defection is serious, using equal penalty for image-tag is too coarse to distinguish missing and noisy tags from other tags, whereas the re-weighted penalty is more reasonable and appropriate to represent the relatedness of pairwise image-tag since the defection can be compensated utilizing both visual similarities and tag correlations.

Fig. 4 demonstrates tag completion results of exemplary images using proposed methods with various degree of incomplete and noisy conditions. It can be learnt from Fig. 4 that using proposed re-weighting scheme, proposed models are able to improve defective tags which are initially as-

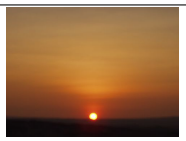
ESP Game	IAPRTC-12	MIR Flickr	ESP Game	IAPRTC-12	MIR Flickr
					
Ground truth : boat, coin, money, old, silver	**Ground truth :** horizon, orange, sky, sun	**Ground truth :** sunset, italy, italia, roma, rome	**Ground truth :** grass, green, house, sky, tree	**Ground truth :** building, car, street	**Ground truth :** yellow, flowers, orange
Incompletion 70%: boat, money	**Incompletion 90%:** sun	**Incompletion 50%:** sunset, roma	**Noise 90%:** bikini, cat, game, green, house, light, sky, tree, grass	**Noise 70%:** building, car, field, shirt, street	**Noise 30%:** yellow, flowers, orange, newyork
Fasttag: coin, metal, head, old, round	**TMC:** cloud, sun, orange, painting, sunset	**Fasttag:** sunset, sky, blue, england, olympus	**TMC:** sand, boat, forest, tree, green	**Fasttag:** field, street, bay, car, road	**TMC:** light, flowers, nyc, yellow, holland
Fasttag-RW: coin(0.13), money(0.91), old(0.32), round(0.45), silver(0.25)	**TMC-RW:** sunset(0.11), horizon(0.35), sun(0.89), orange(0.27), landscape(0.37)	**Fasttag-RW:** sunset(0.88), italy(0.16), italia(0.21), roma(0.83), cityscape (0.35)	**TMC-RW:** grass(0.97), green(0.95), house(0.72), man(0.38), tree(0.81)	**Fasttag-RW:** building(0.95), car(0.81), pavement(0.46), sky(0.41), street (0.8)	**TMC-RW:** macro (0.24), flower(0.16), yellow(0.85), flowers(0.96), orange(0.79)

Fig. 4. Tag completion results of exemplar images on three datasets with various degree of incomplete (left three) and noisy (right three) settings. The fourth row is initial defective tags, last two rows are predicted tags from tag completion models. For proposed models (last row), weighted penalty $Z_{i,j}$ of predicted tags are also presented.

signed, recall related missing tags and reject existed noisy tags. Moreover, the weighted penalties for predicted tags accord to the principle of proposed re-weighting scheme, which allocates smaller penalties to both missing and noisy tags than correctly labeled tags.

5. CONCLUSION

In this paper we propose an image-tag re-weighting scheme and formulate a re-weighted empirical loss term which is more efficient for optimization in tag completion problem under defective tag assignments. In particular, the weighted penalty of pairwise image-tag is derived from image-image similarities and tag-tag associations, which ensures that missing and noisy tags have discriminant importance rather than identical importance in previous works. Experimental evaluations verify that using proposed re-weighting scheme achieves superior tag completion results on various degree of incomplete and noisy settings. In addition, we believe proposed re-weighting scheme can also benefit image annotation performance which assumes complete tag assignments.

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